VLSI DESIGN INTERNSHIP

TASK 2 REPORT

Introduction to Verilog HDL and RTL Design of Basic Combinational Circuits

Submitted by

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1. Objective

To understand Verilog HDL and RTL design, model basic combinational circuits using Verilog, develop testbenches, perform simulation, and analyze the resulting waveforms.

2. Introduction

Verilog HDL is used to describe digital hardware circuits. In this task, Verilog was used to model combinational logic circuits and verify them through testbenches and simulation waveforms.

The Task 2 guide covers Verilog fundamentals, RTL design concepts, six basic logic gates, Half Adder, Full Adder, testbench writing, simulation, and waveform analysis.

3. Concepts Studied

- Verilog syntax and module structure
- Inputs, outputs and wires
- Continuous assignment using assign
- Verilog logic operators
- RTL and combinational logic
- Half Adder and Full Adder
- Sum and Carry logic
- Testbench structure and test inputs
- Simulation flow and waveform interpretation

4. Tools Used

Tool	**Purpose**
EDA Playground	
	Online Verilog editor and simulator
Icarus Verilog	
	Compilation and simulation
EPWave / waveform viewer	
	Viewing and analyzing simulation waveforms

These tools are consistent with the free tools listed in the Task 2 guide.

5. Basic Logic Gates

The following six logic gates were implemented using Verilog and verified using testbenches and simulation waveforms.

5.1 AND Gate

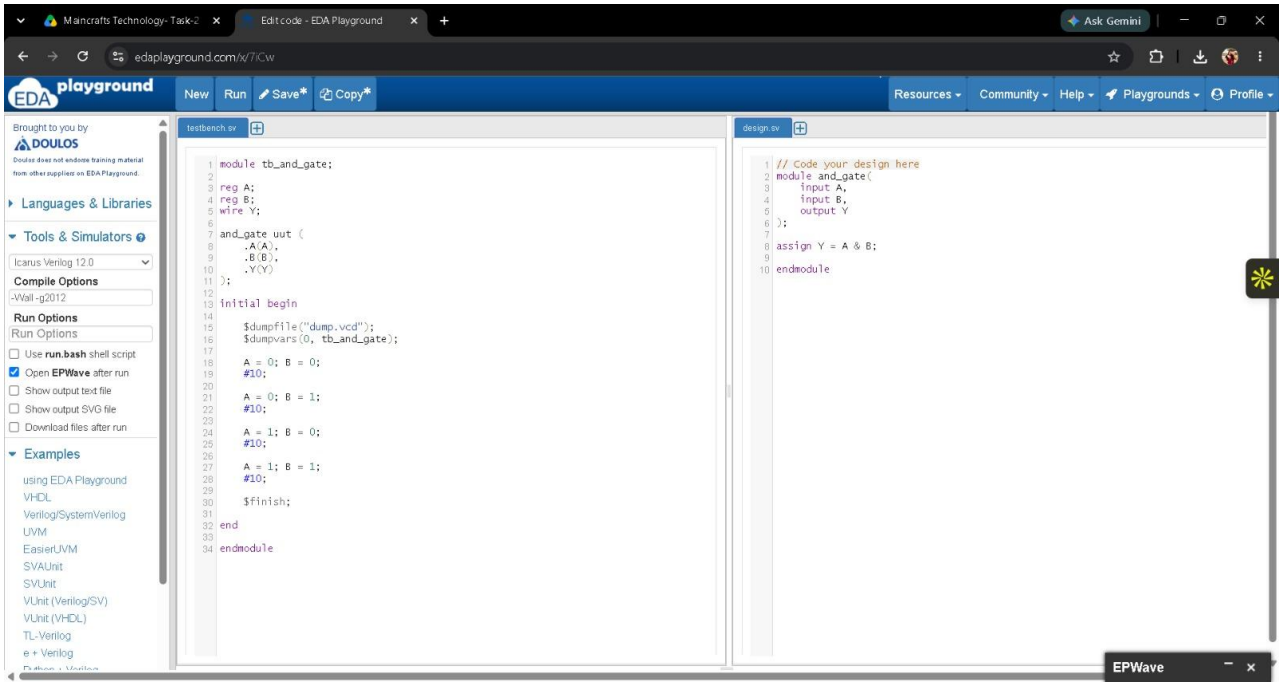
Logic expression: $Y = A \cdot B$

Truth Table

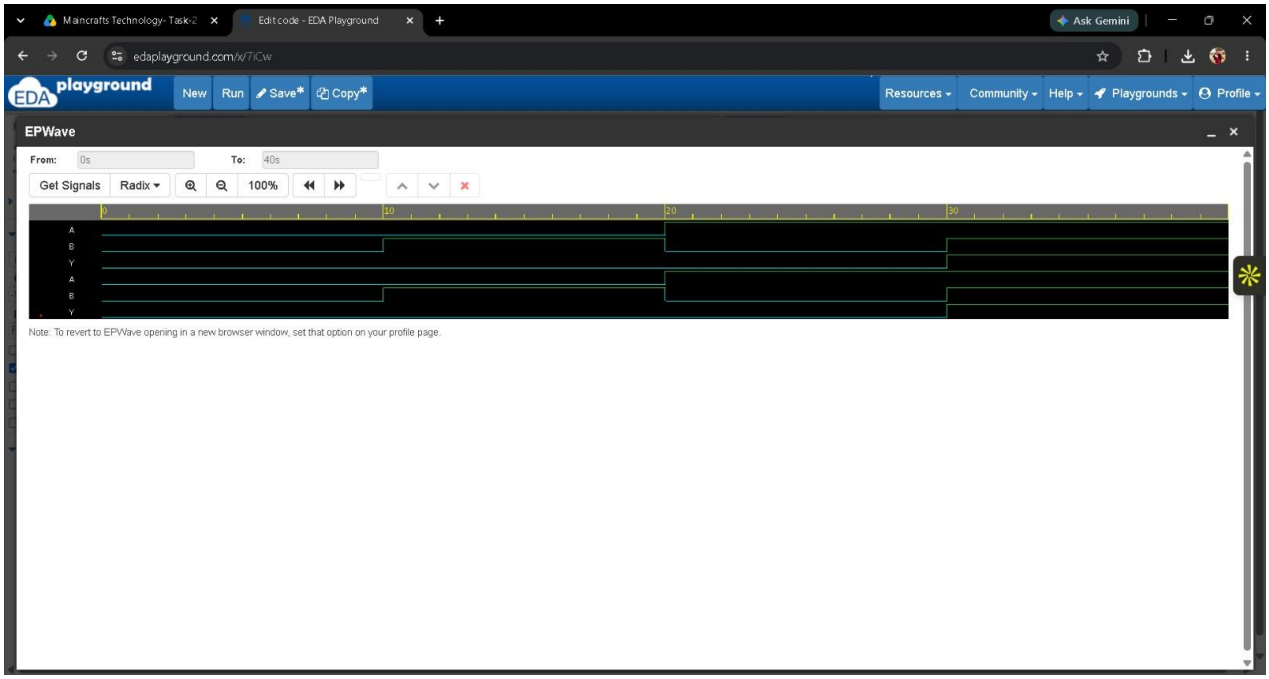
A	**B**	**Y**

0		
	0	
		0
0		
	1	
		0
1		
	0	
		0
1		
	1	
		1

Design and Testbench Code



Simulation Waveform



Observation: The output is 1 only when both inputs are 1.

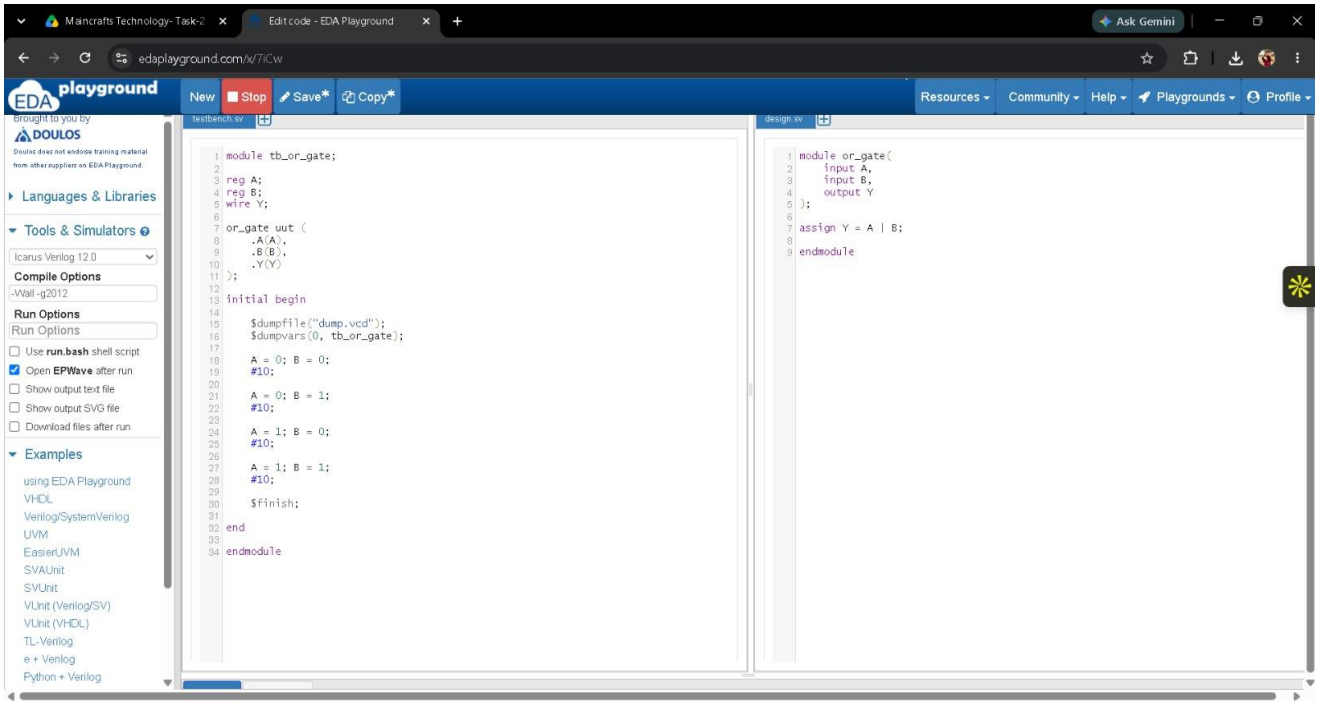
5.2 OR Gate

Logic expression: Y = A + B

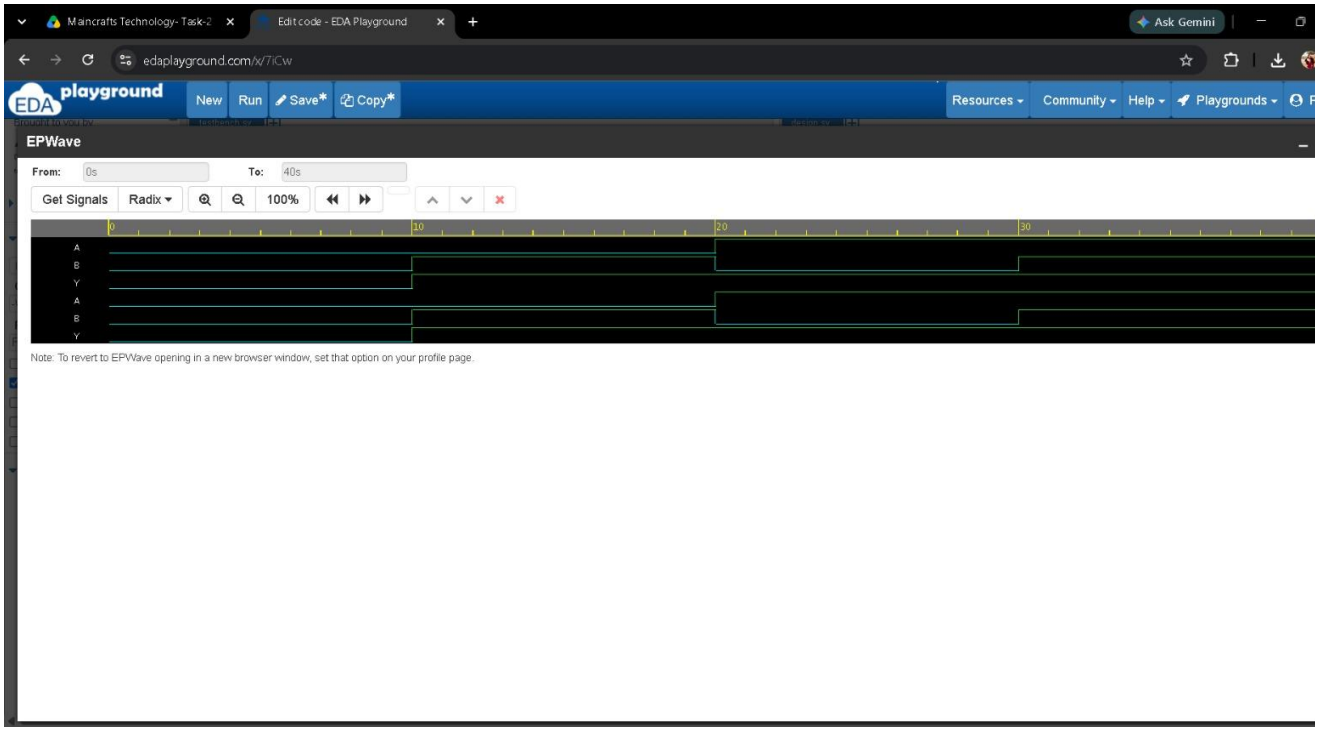
Truth Table

A	B	Y
0		
	0	
		0
0		
	1	
		1
1		
	0	
		1
1		
	1	
		1

Design and Testbench Code



Simulation Waveform



Observation: The output is 1 when at least one input is 1.

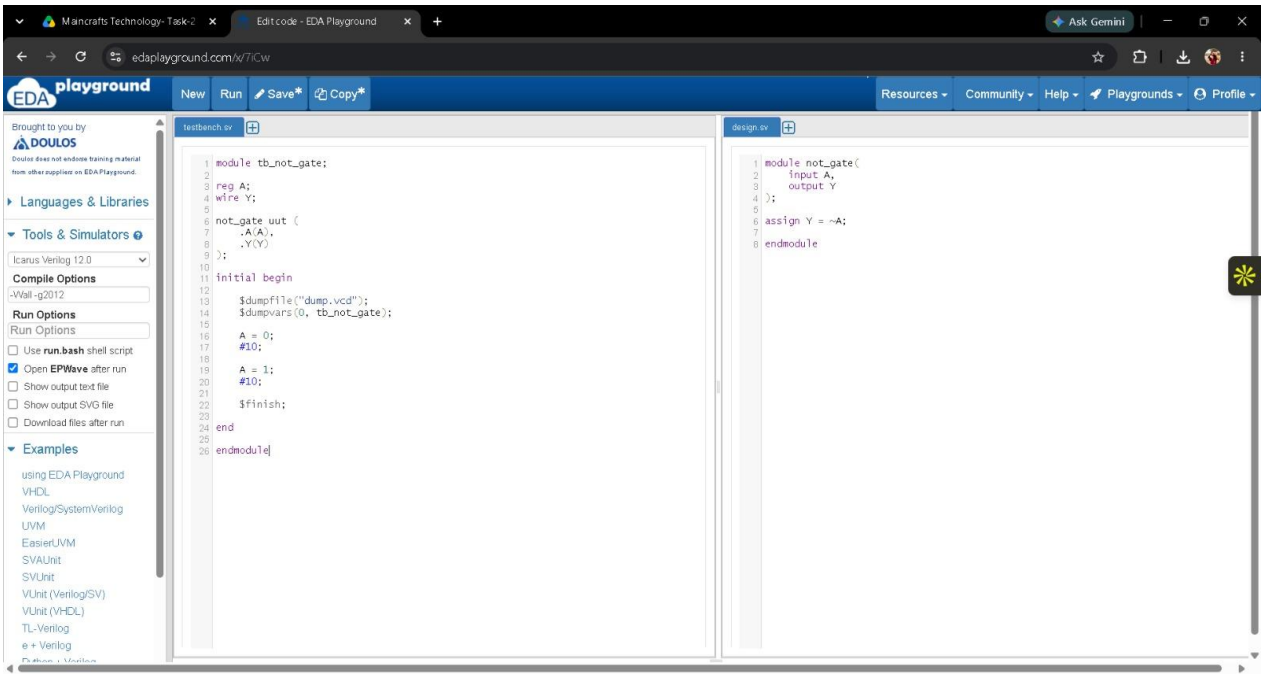
5.3 NOT Gate

Logic expression: $Y = \overline{A}$

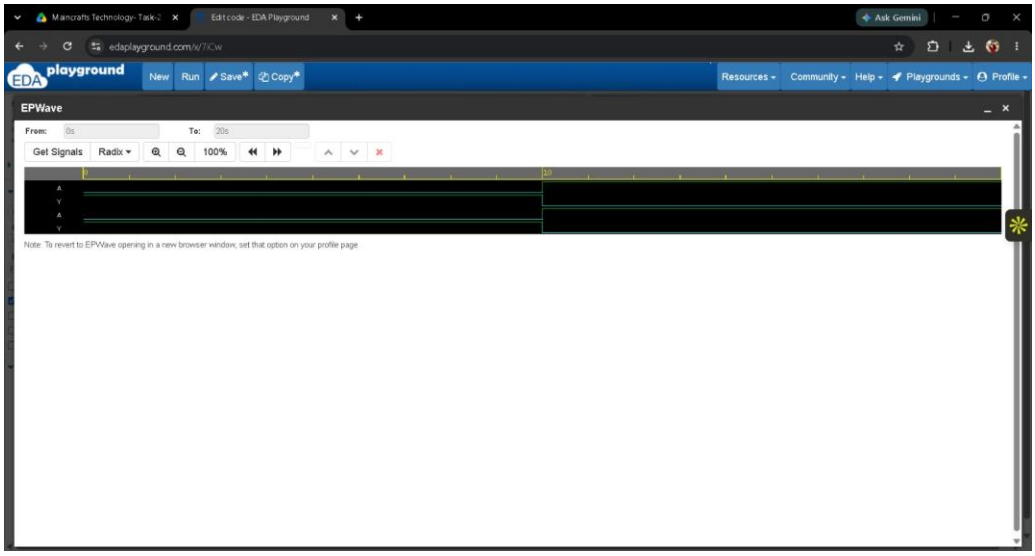
Truth Table

A	Y
0	
	1
1	
	0

Design and Testbench Code



Simulation Waveform



Observation: The output is the complement of the input.

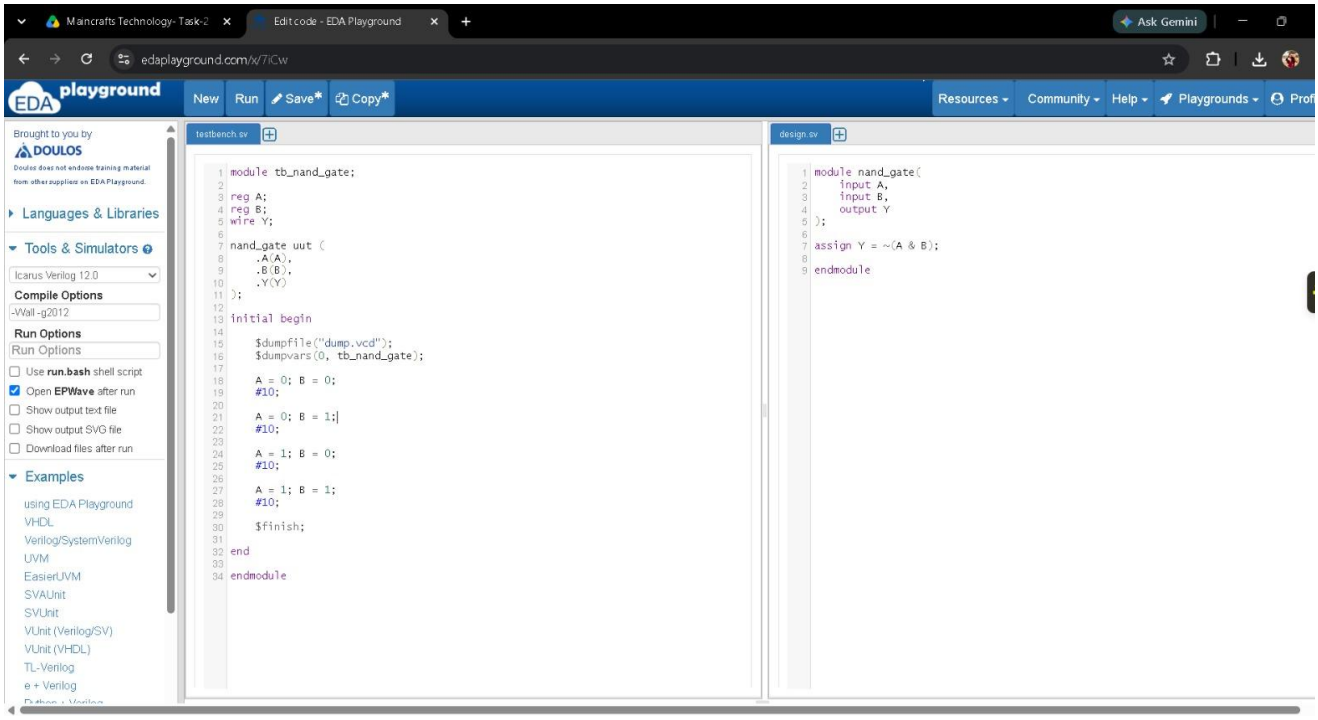
5.4 NAND Gate

Logic expression: $Y = \overline{(A \cdot B)}$

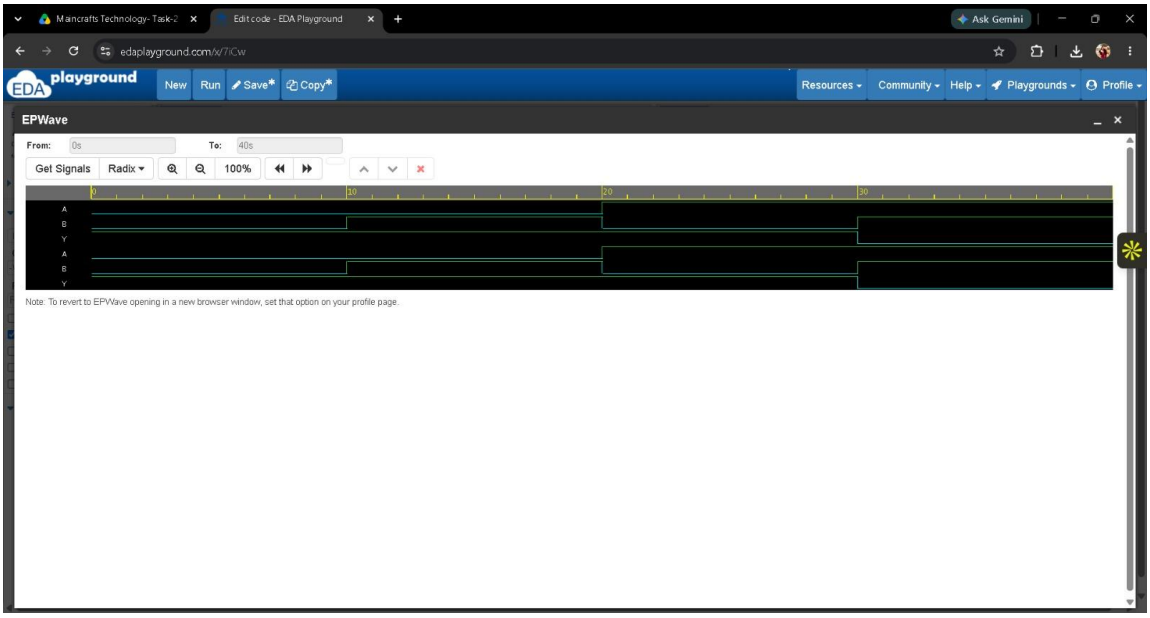
Truth Table

A	B	Y
0		
	0	
		1
0		
	1	
		1
1		
	0	
		1
1		
	1	
		0

Design and Testbench Code



Simulation Waveform



Observation: The output is the complement of the AND operation.

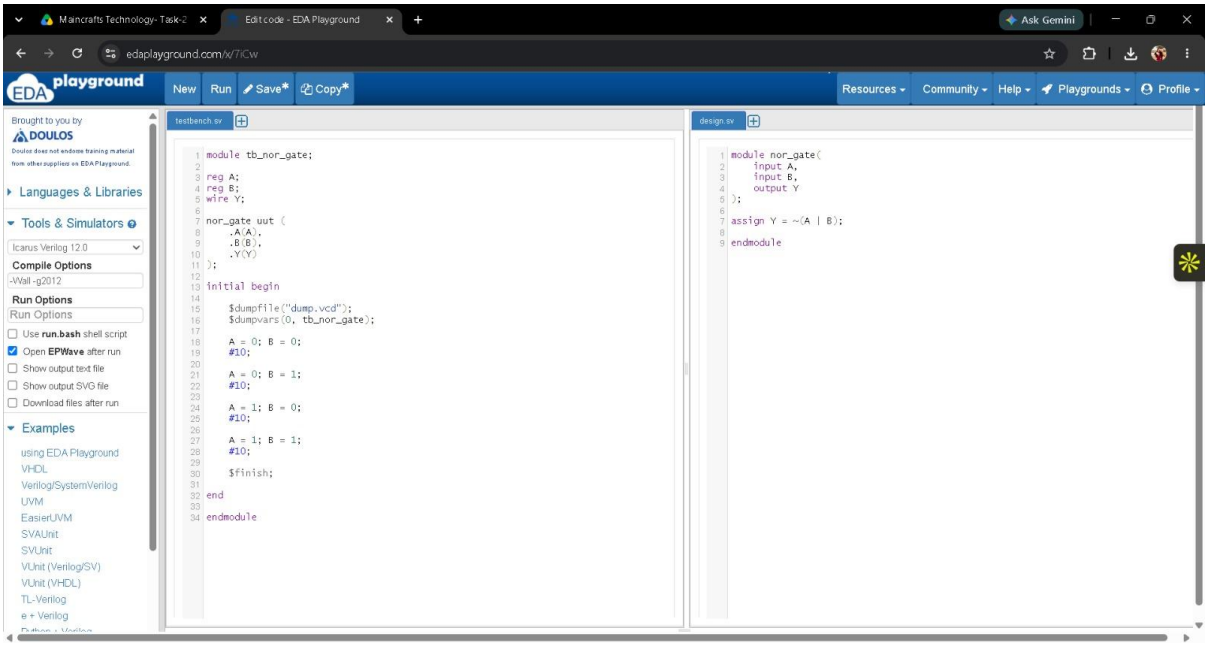
5.5 NOR Gate

Logic expression: $Y = \overline{(A + B)}$

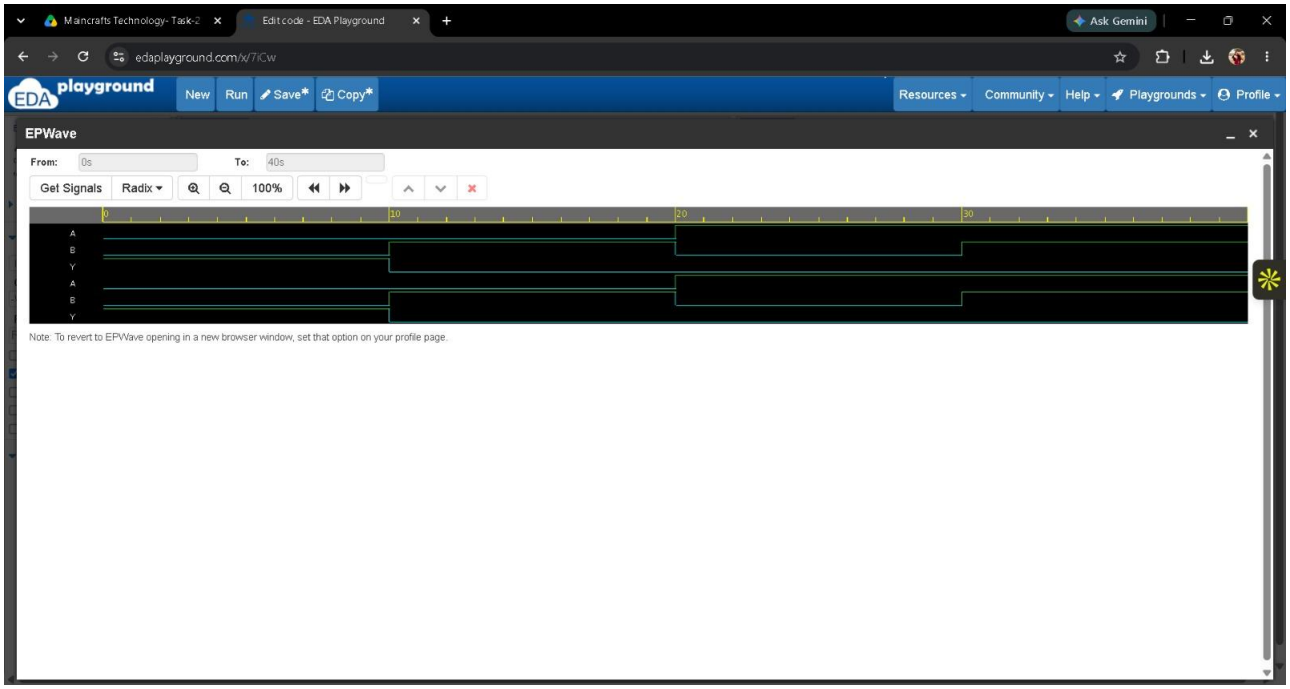
Truth Table

A	B	Y
0		
	0	
		1
0		
	1	
		0
1		
	0	
		0
1		
	1	
		0

Design and Testbench Code



Simulation Waveform



Observation: The output is the complement of the OR operation.

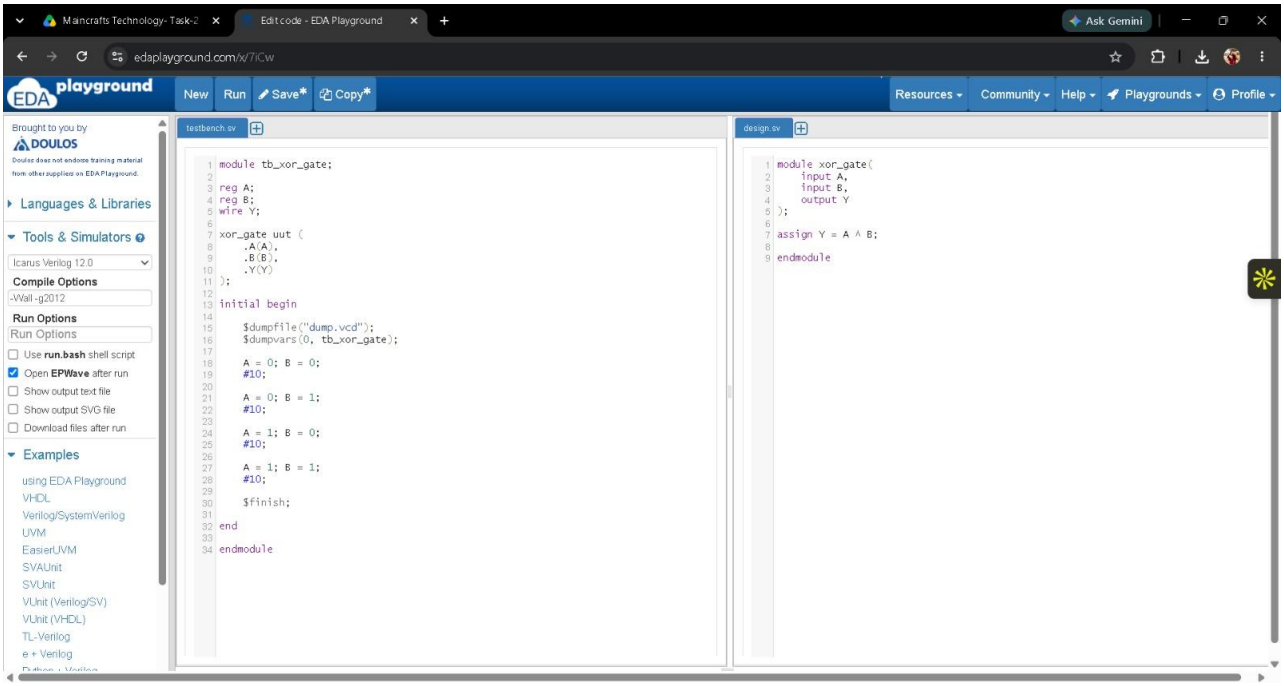
5.6 XOR Gate

Logic expression: $Y = A \oplus B$

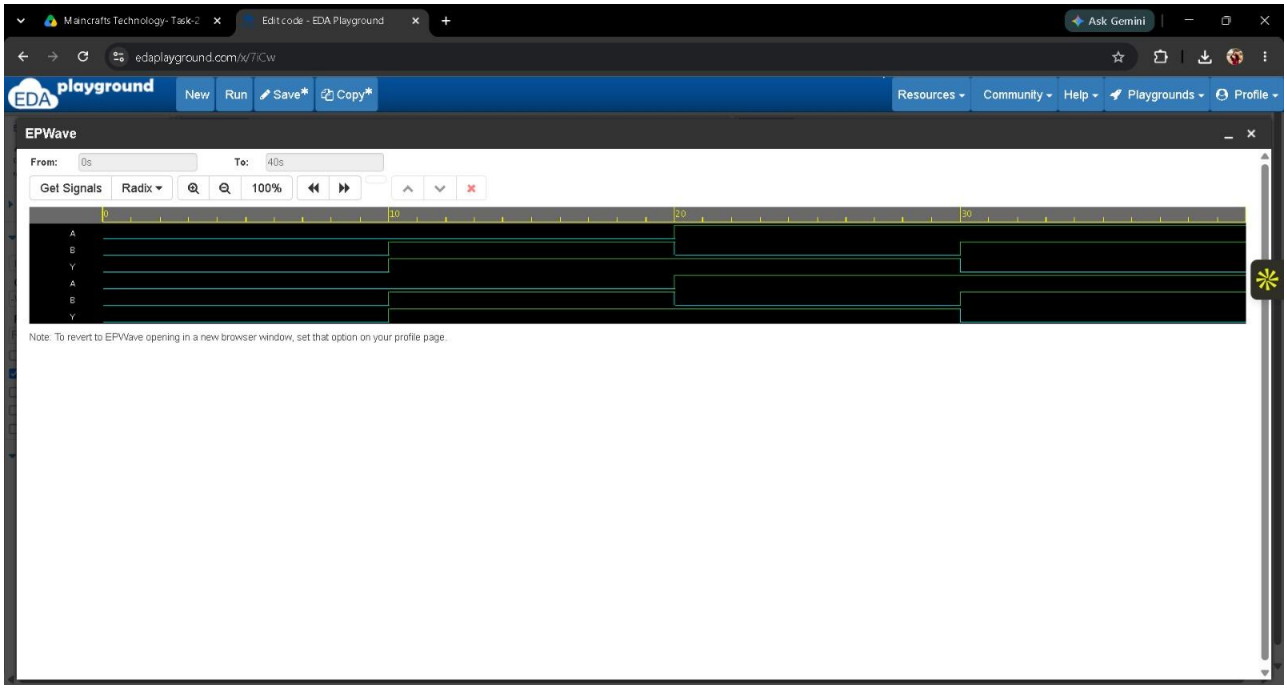
Truth Table

A	B	Y
0		
	0	
		0
0		
	1	
		1
1		
	0	
		1
1		
	1	
		0

Design and Testbench Code



Simulation Waveform



Observation: The output is 1 when the two inputs are different.

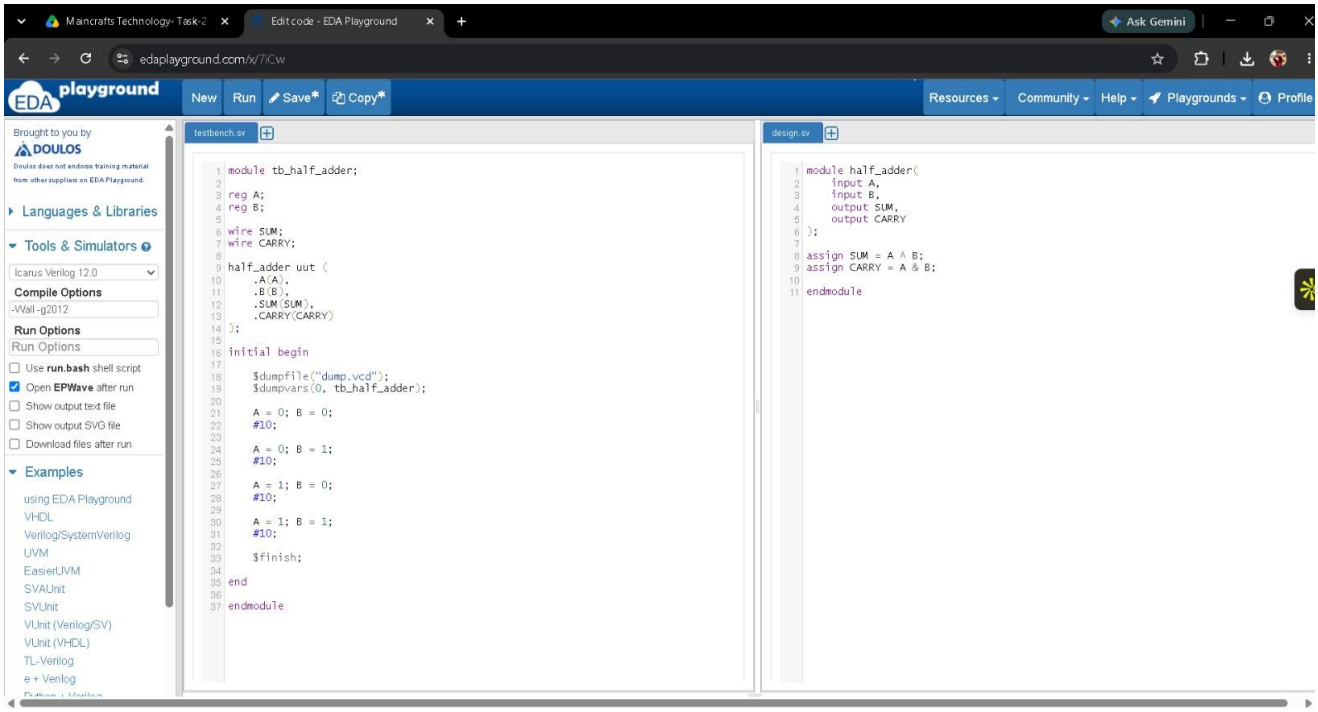
6. Half Adder

A Half Adder is a combinational circuit that adds two one-bit binary inputs and produces SUM and CARRY outputs.

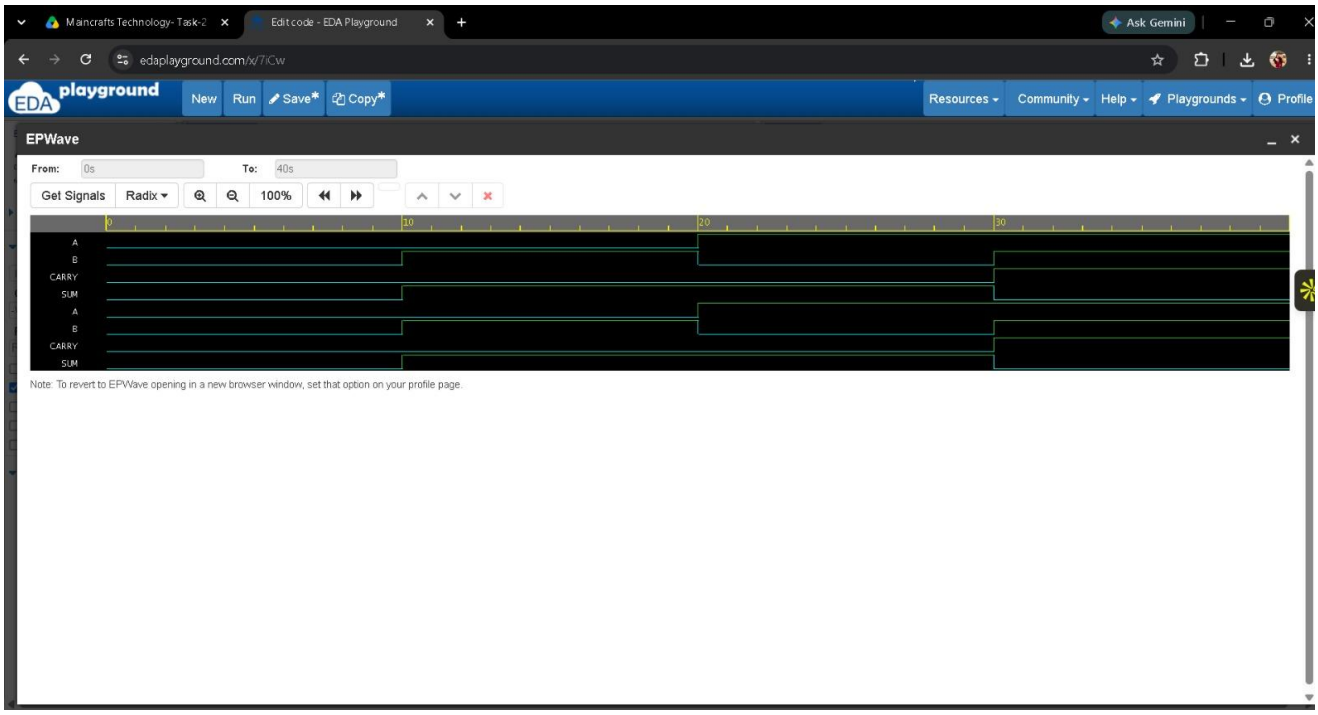
$$\mathbf{SUM = A \oplus B}$$
$$\mathbf{CARRY = A \cdot B}$$

A	B	SUM	CARRY
0			
	0		
		0	
			0
0			
	1		
		1	
			0
1			
	0		
		1	
			0
1			
	1		
		0	
			1

Design and Testbench Code



Simulation Waveform



Observation: The SUM output follows XOR logic and the CARRY output follows AND logic. The simulated outputs were verified against the expected truth table.

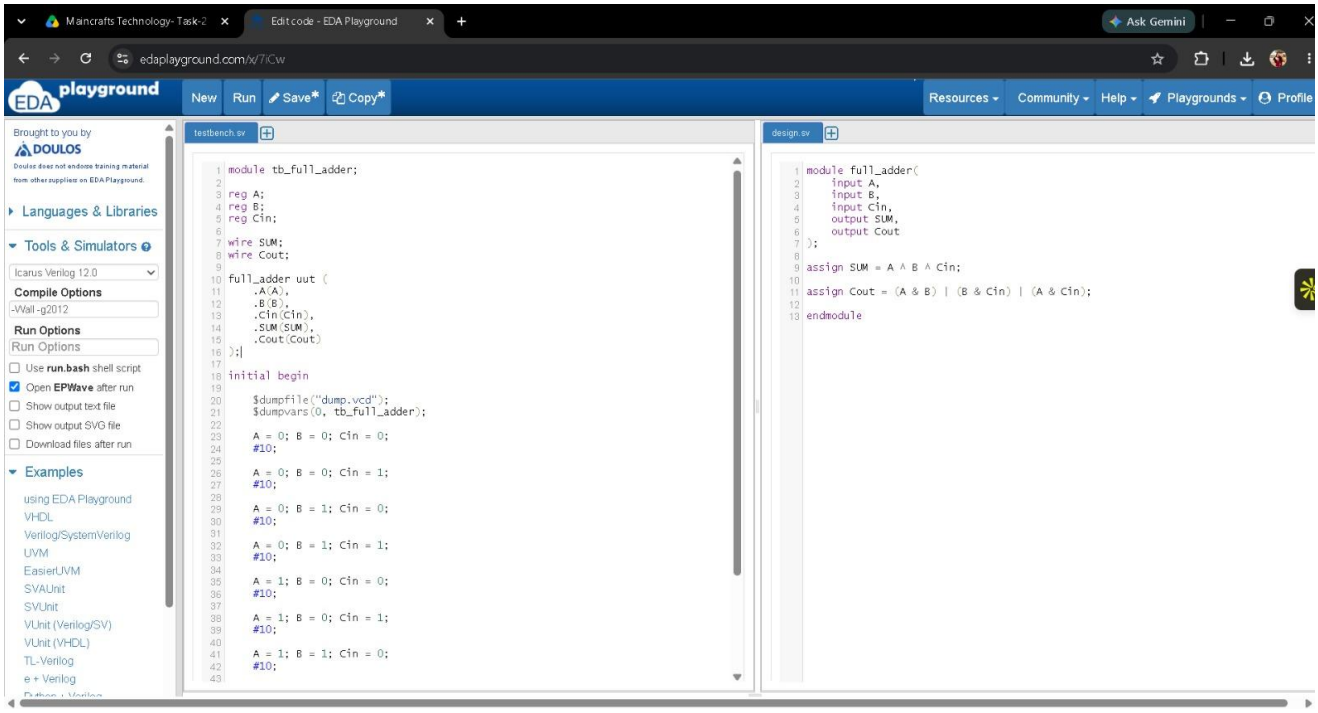
7. Full Adder

A Full Adder is a combinational circuit that adds three one-bit inputs A, B and Cin and produces SUM and Cout outputs.

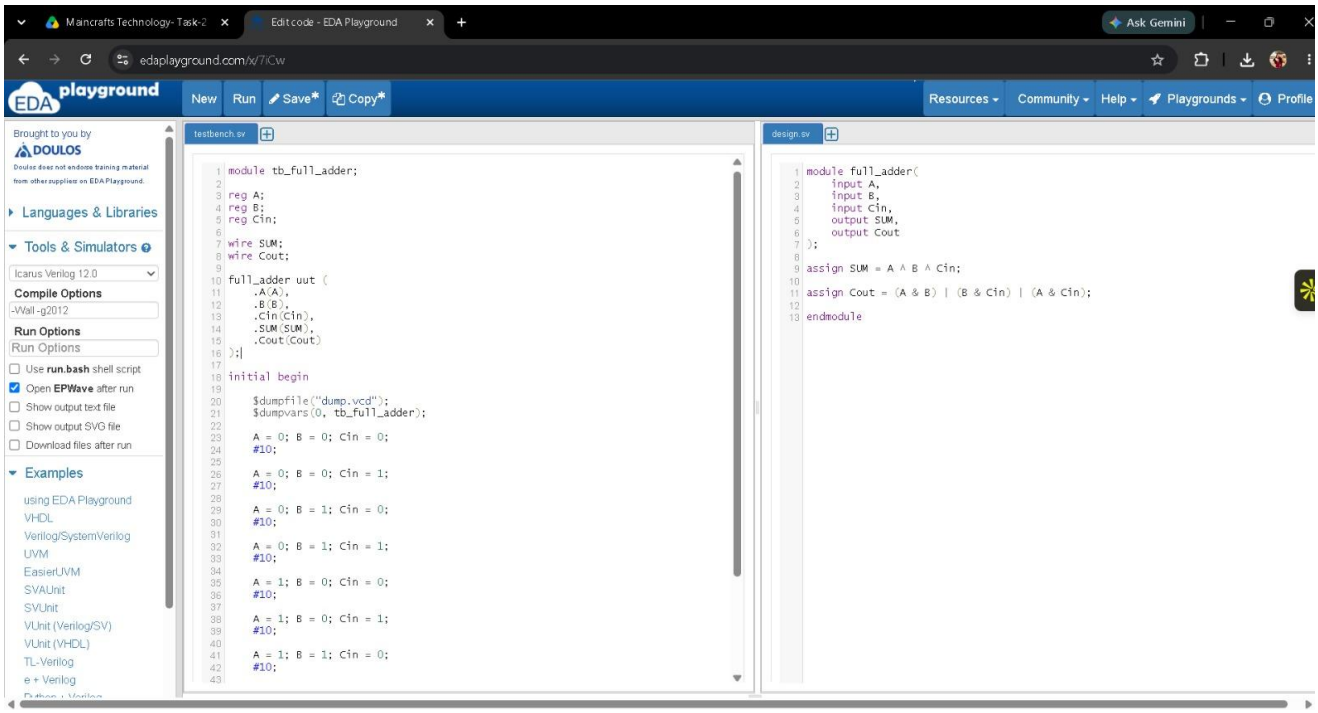
$$\mathbf{SUM = A \oplus B \oplus Cin}$$
$$\mathbf{Cout = (A \cdot B) + (B \cdot Cin) + (A \cdot Cin)}$$

A	B	Cin	SUM	Cout
0				
	0			
		0		
			0	
				0
0				
	0			
		1		
			1	
				0
0				
	1			
		0		
			1	
				0
0				
	1			
		1		
			0	
				1
1				
	0			
		0		
			1	
				0
1				
	0			
		1		
			0	
				1
1				
	1			
		0		
			0	
				1
1				
	1			
		1		
			1	
				1

Design and Testbench Code



Simulation Waveform



Observation: All eight input combinations were tested and the SUM and Cout outputs were verified against the expected truth table.

8. Testbench and Simulation Methodology

A Verilog testbench was used to verify each design. The testbench applies input combinations to the design under test, records simulation activity in a VCD waveform file, and allows the input and output signals to be examined over time.

1. Write the Verilog design module.
2. Write the corresponding testbench.
3. Apply the required input combinations.
4. Run the design and testbench using the simulator.
5. Generate the VCD waveform file.
6. Open the waveform viewer and inspect the signals.
7. Compare simulated outputs with the expected truth table.

9. Waveform Analysis

The waveforms were analyzed by observing changes in input and output signals with simulation time. Four combinations were tested for each two-input gate and Half Adder, while all eight combinations were tested for the Full Adder. The simulated outputs were checked against the expected truth-table behavior.

10. Results

The six basic logic gates, Half Adder and Full Adder were successfully implemented using Verilog HDL. Testbenches were used to apply the required input combinations, and the resulting simulation waveforms were analyzed. The observed outputs matched the expected logic behavior.

11. Learning Outcomes

- Understood Verilog module structure and basic syntax.
- Learned to define inputs and outputs and use continuous assignments.
- Applied Verilog operators to model digital logic.
- Understood combinational RTL modeling.
- Implemented and verified Half Adder and Full Adder circuits.
- Learned the purpose and structure of a Verilog testbench.
- Gained practical experience in simulation and waveform analysis.

12. Challenges and Resolution

During simulation, attention was required to ensure that module names, signal connections and waveform dump scopes matched correctly. These were checked and corrected during simulation until the expected waveforms were obtained.

13. Conclusion

This task provided practical exposure to Verilog HDL and RTL-based modeling of combinational circuits. Implementing logic gates, Half Adder and Full Adder reinforced digital logic concepts, while testbench development and waveform analysis provided hands-on understanding of functional verification and signal timing.

14. References

- VLSI Design Internship – Task 2 Free Learning Resources & Support Guide, Maincrafts.
- EDA Playground – online Verilog editor and simulator.
- Icarus Verilog – open-source Verilog simulator.
- GTKWave / waveform viewer resources referenced in the Task 2 guide.

15. Final Submission Checklist

☐ AND: code + waveform screenshot inserted

☐ OR: code + waveform screenshot inserted

☐ NOT: code + waveform screenshot inserted

☐ NAND: code + waveform screenshot inserted

☐ NOR: code + waveform screenshot inserted

☐ XOR: code + waveform screenshot inserted

☐ Half Adder: code + waveform screenshot inserted

☐ Full Adder: code + waveform screenshot inserted

☐ Truth tables checked

☐ Report reviewed and converted to PDF

☐ Final PDF uploaded to the internship submission form